RR-G-700b
September 8, 1965
SUPERSEDING
Int. Fed. Spec. RR-G-00700a (Navy-Ships)
November 28, 1961
(See 6.8)

FEDERAL SPECIFICATION

GRIP; WIRE, STRAND, WIRE ROPE, AND CABLE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers wedge grips for pulling and tautly holding wires, wire strands, wire ropes, and cables during construction, power line installation, haulage, salvage, or assembly work. The grips may be used as permanent fittings. Parallel jaw and eccentric jaw grips which may be used as temporary fittings are covered in RR-G-691.

1.1.1 *Federal specification coverage.* Federal specifications do not cover all varities of the commodity indicated by the titles of the specifications, but include only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Type, classes, and styles.* Wedge grips shall be of the following type, classes, and styles, as specified (see 6.2):

Type III.—Wedge grips.
- Class 1.—Light duty.
  - Style 1.—Hinged-cover housing.
  - Style 2.—Hinged-braces housing.
- Class 2.—Heavy duty.
  - Style 1.—Box housing.
  - Style 2. — Double-hinge housing ("Carpenter stopper").

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposal, form a part of this specification:

Federal Specifications:

QQ-P-416—Plating, Cadmium (Electrodeposited).
QQ-Z-325—Zinc Coating, Electrodeposited, Requirements for.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

FSC 5120

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Military Specification:

MIL-H-15424—Hand Tools; Packaging of.

Military Standard:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of wedge grips otherwise conforming to this specification.

3.2 Material. Unless otherwise specified herein, wedge grips shall be made of a carbon or alloy steel of commercial grade. All materials shall be of uniform quality and shall be free of injurious defects. Strength and other physical properties shall be suitable for the purpose intended. Unless specified (see 6.2), the grip housings shall be either cast or forged at the manufacturer's option.

3.3 Design. Wedge grips shall consist essentially of a body housing, one or two floating steel wedges, and rollers or linings for facilitating the sliding and consequently gripping action of the floating wedge(s). One or both wedges as specified (see 6.2) shall be grooved to conform to the lay of the specified cable or rope.

3.4 Capacity. The capacity of the grips shall be such as to readily accept and firmly hold the range of sizes from the largest to the smallest size wire, cable, or wire rope listed in tables I and II for the size number of wedge grip specified.

3.5 Proof load. The wedge grip shall be subjected to the applicable proof loads in accordance with the test procedure in 4.4.1. Each wedge grip shall be capable of transmitting to the wire, cable, or wire rope, the amount of proof load listed in tables I and II for the size number of wedge grip specified. While the full proof load is applied, the various parts of the wedge grip shall display no visible twisting, buckling, or other signs of physical weakness and shall grip the wire cable or wire rope with no slippage. After full proof load is released, the various parts of the wedge grip shall work freely and display no distortion, bending, or other damage.

3.6 Finish. All grips shall be either zinc or cadmium coated at the manufacturer's option, unless the type of coating is specified.

3.6.1 *Zinc-coating.* Zinc-coating shall be done either by hot-dip or electroplating process at the manufacturer's option, unless the process to be used is specified.

3.6.1.1 Hot-dip zinc-coating (galvanizing), when applied, shall be in accordance with good commercial practice.

3.6.1.2 Zinc-electrodepositing, when applied, shall conform to type I, class 3 of QQ-Z-325.

3.6.2 Cadmium plating, when applied, shall conform to type I, class C of QQ-P-416.

3.6.3 Unless otherwise specified (see 6.2), electrodeposited zinc or cadmium coatings shall be not less than 0.0002 inch thick.

3.7 Branding. Wedge grips shall be plainly branded or carry a metal plate giving the

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Table I - Class 1, Styles 1 and 2, Light Duty, Wedged Cable Grips.

Size Number	Diameter of Wire Rope or Cable	Weight (Nominal) Pounds	Maximum Safe Load Pounds	Proof Load Pounds
1	3/16	1-1/2	1500	1725
2	1/4	2-1/4	1750	2250
3	5/16	3	2250	2588
4	3/8	3-1/4	4500	5775
5	7/16	4-3/4	6250	7100
6	1/2	6-1/4	8000	9200
7	9/16	8	9250	10400
8	5/8	12	10750	11500
9	3/4	14	12600	14250
10	7/8	15-1/2	13750	15500
11	1	17	15000	17250
12	1-1/8	22	17250	19300
13	1-1/4	25	20000	23000
14	1-3/8	29	25000	28750
15	1-1/2	29	25000	28750

Table II - Class 2, Styles 1 and 2, Heavy Duty, Wedged Cable Grips.

Size Number	Diameter of Wire Rope or Cable	Weight (Nominal) Pounds	Maximum Safe Load Pounds	Proof Load Pounds
20	3/4	21	16300	18250
21	7/8	22-3/4	17650	19700
22	1	25-1/2	21250	24750
23	1-1/8	33	23125	27300
24	1-1/4	38	29700	32150
25	1-3/8	43-1/2	34125	37750
26	1-1/2	45	37300	39125
27	1-5/8	50	40000	46000
28	1-3/4	53-1/2	43175	48125
29	1-7/8	55	47300	50125
30	2	57	49125	53250
31	2-1/8	59	52175	59200
32	2-1/4	61	61300	65750
33	2-1/2	63	67200	72300
34	2-3/4	65	71600	79125
35	3	67	80250	87350

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material, construction and diameter of the largest wire cable or rope to be handled, the figure of the permissible working load, the manufacturer's name or trademark, and the manufacturer's serial number.

3.8 **Weight.** The maximum allowable weight of the wedge grip shall not exceed the nominal weights listed in tables I and II for the size number of wedge grip specified by more than 10 percent.

3.9 **Class 1, light duty.** Class 1 wedge grips shall be either style 1 (see figures 1 and 2) or style 2 (see figure 3), as specified (see 6.2).

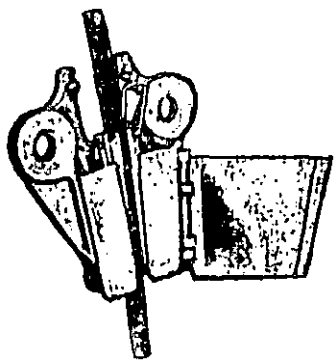


FIGURE 1.—Grip with a double-eye housing, class 1, style 1.

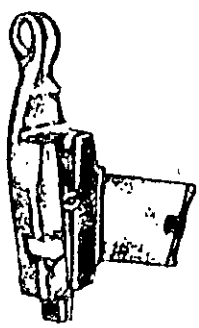


FIGURE 2.—Grip with a single-eye housing, class 1, style 1.

3.9.1 *Style 1, with a hinge-cover housing.* Style 1 wedge grips shall be used on wire cables or ropes from 3/16 to 1 inch in diameter. The housings in the wedge grips shall be of a box shape with a hinged cover, hasp held, for quick and easy insertion or removal of the cable or wire rope and for changing the floating gripping wedges to suit the cable or wire rope to be handled. The hasp shall be pin locking for security. The wedge ways shall be equipped with rollers or with bronze linings at the manufacturer's option, or as specified (see 6.2). The housing shall be either of double-eye type (see figure 1), or single-eye type (see figure 2) for attaching the grip to a two or a single-leg sling as specified (see 6.2). Each grip shall be made for use of two floating gripping wedges. Grips intended for handling a specific wire cable or rope shall have both wedges grooved to fit its lay, whereas, grips intended for use on two or more sizes and constructions of wire cables or ropes shall have one floating wedge with a smooth V-shape groove and the complementing floating wedges shall be grooved with the lays of the specified wire cables or ropes. Style 1 grips shall conform to table I.



FIGURE 3.—Grip with hinged-braces housing (and bridles), class 1, style 2.

3.9.1.2 *Style 2, with hinged-braces housings.* Style 2 wedge grips shall be used on

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FIGURE 4.—Grip with a box housing, class 2, style 1.

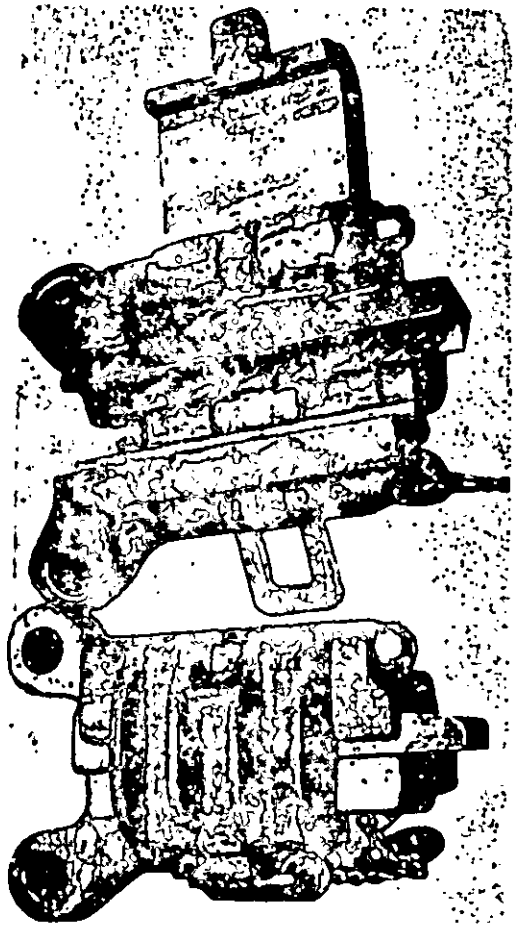


FIGURE 5.—Grip with a double-hinge housing ("carpenter stopper"), class 2, style 2.

wire cables or ropes from 1/2 to 1-1/2 inches in diameter. The housing in the wedge grips shall be of a box shape, with two eyes for attaching it to a sling, and instead of a cover shall have two hinged braces, pin locking. Style 2 grips shall meet the requirements specified for grips of the style 1 (see 3.9.1), shall be similar to figure 3 and shall conform to table I.

3.9.2 *Class 2, heavy duty.* Class 2 wedge grips shall be either style 1, (see figure 4) or style 2, (see figure 5), as specified (see 6.2).

3.9.2.1 *Style 1, with box housings.* Style 1 wedge grips shall be used on wire cables and ropes from 3/4 to 3 inches in diameter. The housing in these grips shall be of tapered box shape heavily cross-ribbed on all sides. The cover shall be removable (bolted for insertion, removal or changing of the floating wedges to suit the wire cables or ropes to be handled). The wedge ways shall be equipped with rollers or bronze lined at the manufacturer's option, or as specified (see 6.2). In large-size grips the floating

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wedges shall be spring closed for holding the inserted wire cable or rope until the load is applied and shall be interlinked to assure their parallel movement whether free or under a load. Wedge grips for handling one specific wire cable or rope shall have both floating wedges grooved to fit its lay. Wedge grips for handling more than one wire cable or rope shall have one floating wedge with a smooth V-shape groove and each complementing floating wedge grooved with the lay of the wire cable or rope to be used. Style 1 grips shall be similar to figure 4 and shall conform to table II.

3.9.2.2 *Style 2, with double-hinge housings.* Style 2 wedge grips shall be used on wire ropes and cables from 3/4 to 3 inches in diameter. The housing in these grips shall consist of three hinged sections. When closed and locked by a hasp, the sections shall form a rigid grip body, in which one side has an inclined V-shape smooth groove while the other has ways for a sliding wedge. The ways to be bronze lined at the manufacturer's option, or as specified (see 6.2). The sliding wedge shall be grooved with a lay of the wire cable or rope to be handled. Style 2 grips shall be similar to figure 5 and shall conform to table II.

3.10 **Bridles.** When specified (see 6.2), the wedge grips shall be supplied with the following bridles for facilitating their handling or storage.

3.10.1 *Bridles for the grip housing.* When specified (see 6.2), the wedge grips shall be supplied with a bridle for facilitating their handling by a single-leg sling. The bridle shall consist of a proper diameter and length wire rope with open sockets on both ends and pin attached to the grip housing (see figures 3 and 4).

3.10.2 *Bridles for wedges.* When specified (see 6.2), floating wedges for grips with two sliding wedges shall be supplied with a bridle interconnecting both wedges for quick and easy removal of them and safe storage without separation. The bridles shall consist of a wire rope of proper diameter and length, socketed and pin attached to eye screws in the wedges (see figures 3 and 4).

3.11 **Workmanship.** The workmanship shall be compatible with the type, class, and style of the grips specified. The grips shall be free from fins, burrs, slivers, rust, scale, sharp edges and other defects which may affect their serviceability or handling. No welding or brazing of the grip body or housing shall be permitted. Coating shall be adherent, free from lumps, blisters, uncoated spots and other defects which may affect their use or the protection to the base metal. All assembled parts shall fit properly assuring ease of operation.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for quality conformance inspection.** Sampling shall be performed in accordance with the provisions set forth in MIL-STD-105 except where otherwise indicated herein.

4.3 **Inspection of the end item.** Inspection of the grips shall be performed as specified in the following paragraphs. A lot shall consist of grips of the same type, class and style. The sample unit shall be one completely fabricated grip

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4.3.1 *Visual examination.* Each sample unit shall be visually examined for the defects set forth below. The inspection level shall be level II with an acceptable quality level of 6.5 defects per hundred units.

Examine	Defects
Finish	Not zinc or cadmium coated. Burnt, black, or streaky deposit. Thin spots or uncoated areas. Peeling, chipping, pits, or nodules.
Material	Not fabricated from specified material.
Design	Not type, class, or style specified.
Construction and workmanship	Component missing, broken, warped, or otherwise defective. Welding or brazing used on grip body or housing. Operation omitted or not properly performed. Components do not fit properly, hindering operation. Wedges not grooved for lay of wire as specified.
Marking	Missing, incomplete, or not legible.

4.3.2 *Dimensional examination.* Each sample unit shall be examined for dimensions specified in applicable paragraphs. The inspection level shall be level I with an acceptable quality level of 4.0 defects per hundred units.

4.3.3 *Testing.* Each sample unit shall be tested as specified in 4.4. The inspection level shall be I*A* with an acceptable quality level of 1.5 defects per hundred units.

4.3.4 Inspection of preparation for delivery shall be performed in accordance with MIL-H-15424.

4.4 Test procedures.

4.4.1 *Proof-load test.* Grips shall be subjected to a tensile test to determine compliance with 3.5. The maximum speed of jaw separation of the tensile-testing device shall not exceed a rate of 1 foot per minute for the test. Grips shall be tested using the type of wire, cable, or wire rope for which the particular grip under test was designed. In the event that the tensile strength of any specific type of wire, cable, or wire rope be less than the proof load specified for the cable grip being tested, a wire, cable, or steel rod which will withstand the proof load specified shall be substituted for test purposes.

4.4.2 *Tests of material and of coating.* Tests of the material or of the coating in the grips are not required unless such tests are specified in the contract or order. The tests shall conform to the requirements stated in the contract or order, or in the applicable specifications listed in 2.1.

5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 Cleaning, preservation, and packaging.

5.1.1 *Level A.* Level A cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424.

5.1.2 *Level C.* Level C cleaning, preservation, and packaging shall be in accordance with the supplier's commercial practice.

5.2 Packing.

5.2.1 *Level A.* Level A packing shall be in accordance with level A of MIL-H-15424.

5.2.2 *Level B.* Level B packing shall be in accordance with level B of MIL-H-15424.

5.2.3 *Level C.* Wedge grips preserved and packaged as specified (see 6.2), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with rules or regulations as applicable to the mode of transportation.

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5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior and exterior containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military activities.* Marking shall be in accordance with MIL-H-15424.

6. NOTES

6.1 Intended use. Wedge grips covered by this specification are intended for use in construction, bridge building, heavy hauling, towing and salvage work. Permissible working load on these grips shall not exceed 1/5th of the breaking (maximum) strength of the wire rope or cable to be used.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Class and style required (1.2).
(*c*) When the composition of the steel required for grip housings is different than specified (3.2).
(*d*) When the grip housings required shall not be cast or forged (3.2).
(*e*) Whether one or both wedges shall be grooved (3.3).
(*f*) Type of coating, if other than specified (3.6).
(*g*) Process to be used for zinc coating if different (3.6.1).
(*h*) When thickness of zinc or cadmium coating is different (3.6.3).
(*i*) When wedge ways shall be other than equipped with rollers or bronze lined (3.9.1, 3.9.1.2, and 3.9.2.1).
(*j*) Whether housing for style 1 grip shall be single-eye or double-eye type (3.9.1).
(*k*) When wedge grips shall be supplied with bridles (3.10).
(*l*) Tests required (4.4)
(*m*) Applicable level of preservation, packaging, and packing required (5.1 and 5.2).

6.3 Types I and II grips are covered by Federal Specification RR-C-691a, dated May 1, 1963.

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:
Tools, not otherwise indexed by name. Carload minimum weight—30,000 pounds.

Motor:
Tools, hand, not otherwise indexed. Truckload minimum weight—30,000 pounds, subject to Rule 115, National Motor Freight Classification.

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